Course Type	Course Code	Course Name	Credits
OEC	OEC7015	Operation Research	03

Examination Scheme					
Distribution of Marks			**Exam Duration (Hrs.)**		**Total Marks**
In-semester Assessment		**End Semester Exam (ESE)**			
Continuous Assessment	**Mid-Semester Exam (MSE)**		**MSE**	**ESE**	
20	30	50	1.5	2	100

Program Outcomes addressed:

1. PO1: Engineering knowledge
2. PO2: Problem analysis
3. PO3: Design/development of solutions
4. PO6: The Engineer & The world
5. PO10: Project Management & Finance
6. PO11: Lifelong Learning

Course Objectives:

1. To introduce the concept of formulation of a real-world problem as a mathematical programming model.
2. To impart the knowledge of the mathematical tools that are needed to solve optimization problems.
3. To make learner able to use mathematical software to solve the proposed models.
4. To impart the mathematical understanding to analyse and solve integer programming from a wide range of applications.

Module	Detailed Contents	Hrs
	Course Introduction Operations Research (OR) course is an interdisciplinary field that applies advanced analytical, mathematical, and statistical methods to make better decisions in complex systems, optimizing processes and resource allocation. Students learn to identify problems, formulate mathematical models like linear programming, and use algorithms to find optimal solutions for real-world issues in areas such as engineering, management, finance, and logistics.	**01**
01.	**Introduction to Operations Research** **Content:** Introduction, Structure of the Mathematical Model, Limitations of Operations Research **Linear Programming:** Introduction, Linear Programming Problem, Requirements of LPP, Mathematical Formulation of LPP, Graphical method, Simplex Method, Penalty Cost Method or Big M-method, Two Phase Method, Revised simplex method, Duality, Primal – Dual construction, Symmetric and Asymmetric Dual, Weak Duality Theorem, Complimentary Slackness Theorem, Main Duality Theorem, Dual Simplex Method, Sensitivity Analysis Transportation Problem: Formulation, solution, unbalanced Transportation	**15-17**

	problem. Finding basic feasible solutions – Northwest corner rule, least cost method and Vogel's approximation method. Optimality test: the stepping stone method and MODI method. **Assignment Problem:** Introduction, Mathematical Formulation of the Problem, Hungarian Method Algorithm, Processing of n Jobs Through Two Machines and m Machines, Graphical Method of Two Jobs m Machines Problem Routing Problem, Travelling Salesman Problem **Integer Programming Problem:** Introduction, Types of Integer Programming Problems, Gomory's cutting plane Algorithm, Branch and Bound Technique. Introduction to Decomposition algorithms.	
02.	**Queuing models**	**04-06**
	Contents: Queuing systems and structures, single server and multi-server models, Poisson input, exponential service, constant rate service, finite and infinite population.	
03.	**Simulation**	**06-08**
	Contents: Introduction, Methodology of Simulation, Basic Concepts, Simulation Procedure, Application of Simulation Monte-Carlo Method: Introduction, Monte-Carlo Simulation, Applications of Simulation, Advantages of Simulation, Limitations of Simulation	
04.	**Dynamic programming**	**06-08**
	Contents: Characteristics of dynamic programming. Dynamic programming approach for Priority Management employment smoothening, capital budgeting, Stage Coach/Shortest Path, cargo loading and Reliability problems.	
05.	**Game Theory**	**04-06**
	Contents: Competitive games, rectangular game, saddle point, minimax (maximin) method of optimal strategies, value of the game. Solution of games with saddle points, dominance principle. Rectangular games without saddle point – mixed strategy for 2 X 2 games.	
06.	**Inventory Models**	**04-06**
	Contents: Classical EOQ Models, EOQ Model with Price Breaks, EOQ with Shortage, Probabilistic EOQ Model	
	Course Conclusion	**01**
	Total	**45**

Course Outcomes: Learner will be able to

1. Apply simplex method, analyse the relationship between a linear program and its dual, including strong duality and complementary slackness.
2. Perform sensitivity analysis to determine the direction and magnitude of change of a model's optimal solution as the data change
3. Solve specialized linear programming problems like the transportation and assignment problems.

4. Solve network models like the shortest path, minimum spanning tree, and maximum flow problems
5. Analyse the applications of integer programming and a queuing model and compute important performance measures

CO-PO Mapping Table with Correlation Level

CO ID	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
OEC7015.1	3	3	3	-	-	-	-	-	-	-	-
OEC7015.2	3	3	-	-	-	-	-	-	-	-	-
OEC7015.3	3	3	3	-	-	3	-	-	-	3	3
OEC7015.4	3	3	3	-	-	3	-	-	-	3	3
OEC7015.5	3	3	3	-	-	3	-	-	-	-	3
Average	**3**	**3**	**3**	**-**	**-**	**3**	**-**	**-**	**-**	**3**	**3**

Text Books :

1. Operations Research - An Introduction, Taha, H.A., Seventh Edition 2002, Prentice Hall.
2. Operations Research, S. D. Sharma, First Edition, 2012, KedarNath Ram Nath-Meerut Company
3. Operations Research, Kanti Swarup, P. K. Gupta and Man Mohan, Twentieth Edition, 2022 Sultan Chand & Sons

Reference Books :

1. Operation Research-Methods and problems, Saiesini M.,Yaspan A., First Edition, 1959, Prentice Hall
2. Operations Research: Principles and Practice, Ravindran, A, Phillips, D. T and Solberg, J. J., Second edition, 2009, John Willey and Sons.
3. Introduction to Operations Research, Hiller, F. S. and Liebermann, G. J., Fourth edition, 2002, Tata McGraw Hill

Other Resources :

1. NPTEL Course: Introduction to Operation Research, Prof. G.Srinivasa, IIT Madras
   Web Link- https:// nptel.ac.in/courses/110106062
2. NPTEL Course: Operations Research, Prof. Kusum Deep, IIT Roorkee
   Web Link- https://onlinecourses.nptel.ac.in/noc22_ma48/preview

IN-SEMESTER ASSESSMENT (50 MARKS)

1. Continuous Assessment – Theory (20 Marks)

2. Mid Semester Exam (30 Marks)

Mid semester examination will be based on 40% to 50% syllabus.

END SEMESTER EXAMINATION (50 MARKS)

End Semester Examination will be based on syllabus coverage up to the Mid Semester Examination (MSE) carrying 20%-30% weightage, and the syllabus covered from MSE to ESE carrying 70%-80% weightage.

Course Type	Course Code	Course Name	Credits
OEC	OEC7016	Cyber Security and Laws	03

Examination Scheme					
Distribution of Marks			**Exam Duration (Hrs.)**		**Total Marks**
In-semester Assessment		**End Semester Examination (ESE)**			
Continuous Assessment	**Mid-Semester Exam (MSE)**		**MSE**	**ESE**	
20	30	50	1.5	2	100

Program Outcomes addressed:

1. PO 1: Engineering knowledge
2. PO 2: Problem analysis
3. PO 3: Design and Development of Solution
4. PO 4: Conduct investigation of Complex Problem
5. PO 5: Engineering Tool Usage
6. PO 11: Life-long learning

Course Objectives:

1. To define and categorize various forms of cybercrime, including malware attacks, social engineering, financial fraud, and network intrusions, while understanding their technical and societal impacts.
2. To analyse the legal frameworks governing cyber activities, with a focus on distinguishing between different types of cyber laws at national and international levels.
3. To examine the key provisions and structure of the Indian IT Act, 2008, and to evaluate the implications of its latest amendments on digital governance, privacy, and cybercrime adjudication.
4. To identify and compare global information security standards and compliance frameworks, such as ISO 27001, PCI-DSS, GDPR, HIPAA, and NIST, and their relevance to organizational security postures.
5. To apply legal and compliance knowledge in practical scenarios, enabling students to assess organizational compliance requirements and recommend appropriate security controls within legal boundaries.

Module	Details	Hrs.
	Course Introduction This course provides a comprehensive introduction to the principles, practices, and legal frameworks of cyber security. Students will explore various types of cybercrimes, attack methodologies, and defensive strategies, with a strong emphasis on legal and regulatory compliance. The course covers Indian cyber laws and compliances, including the IT Act 2000 and its amendments, as well as global security standards. Through theoretical insights, learners will be equipped to identify, analyse, and respond to cyber threats within a legal and ethical context.	**01**
01.	**Introduction to Cybercrime** **Contents:** Cybercrime definition and origins of the world, Cybercrime and information security, Classifications of cybercrime, Cybercrime and the Indian ITA 2000, A global Perspective on cybercrimes.	**5-6**
02.	**Cyber offenses & Cybercrime** **Contents:** How criminal plan the attacks, Social Engg, Cyber stalking, Cyber café and Cybercrimes, Botnets, Attack vector, Cloud computing, Proliferation of Mobile and Wireless Devices, Trends in Mobility, Credit Card Frauds in Mobile and Wireless Computing Era, Security Challenges Posed by Mobile Devices, Registry Settings for Mobile Devices, Authentication Service Security, Attacks on Mobile/Cell Phones, Mobile Devices: Security Implications for Organizations, Organizational Measures for Handling Mobile, Devices-Related Security Issues, Organizational Security Policies and Measures in Mobile Computing Era, Laptops.	**7-8**
03.	**Tools and Methods Used in Cyberspace** **Contents:** Phishing, Password Cracking, Key loggers and Spywares, Virus and Worms, Steganography, DoS and DDoS Attacks, SQL Injection, Buffer Over Flow, Attacks on Wireless Networks, Phishing, Identity Theft (ID Theft)	**7-8**
04	**The Concept of Cyberspace** **Contents:** The Concept of Cyberspace, E-Commerce, The Contract Aspects in Cyber Law, The Security Aspect of Cyber Law, The Intellectual Property Aspect in Cyber Law, The Evidence Aspect in Cyber Law, The Criminal Aspect in Cyber Law, Global Trends in Cyber Law, Legal Framework for Electronic Data Interchange, Law Relating to Electronic Banking, The Need for an Indian Cyber Law.	**7-8**
05	**Indian IT Act** **Contents:** Cyber Crime and Criminal Justice, Penalties under the IT Act, 2000, Adjudication Mechanism, Appeals under the IT Act, Information Technology Act, 2000, IT Act, 2008 and its Amendments.	**5-6**
06	**Information Security Standard compliances** **Contents:** SOX, GLBA, HIPAA, ISO, FISMA, NERC, PCI.	**4-5**
	Course Conclusion	

	This course equips learners with essential knowledge of cybercrime, cybersecurity practices, and cyber laws governing the digital ecosystem. Students gain an understanding of the Indian IT Act and its amendments, various aspects of cyber law, cybercrime investigation and adjudication processes, and major information security compliance standards. Overall, the course enables learners to analyze cyber threats, interpret legal and regulatory requirements, and apply security and compliance principles for secure and lawful use of information systems.	**01**
	Total	**45**

Course Outcomes: A learner will be able to –

1. Classify various types of cybercrimes, cyber threats, and attack techniques, and analyze their technical and societal impacts in digital environments.
2. Analyze cyberattack scenarios and security challenges to identify vulnerabilities, attack vectors, and appropriate technical and organizational security measures.
3. Interpret and apply provisions of the Indian Information Technology Act, 2000 and its amendments to address cybercrime investigation, adjudication, and legal issues.
4. Analyze different aspects of cyber law, including cyberspace governance, e-commerce, digital contracts, intellectual property, electronic evidence, and criminal liability.
5. Apply cybersecurity, legal, and compliance knowledge to real-world scenarios to recommend lawful, secure, and compliant practices in the design and use of information systems.

CO-PO Mapping Table with Correlation Level

CO ID	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
OCE7016.1	3	-	-	-	-	-	-	-	-	-	-
OCE7016.2	3	3	3	3	3	-	-	-	-	-	3
OCE7016.3	3	3	-	-	-	-	-	-	-	-	-
OCE7016.4	3	3	-	-	-	-	-	-	-	-	-
OCE7016.5	3	3	-	-	-	-	-	-	-	-	-
Average	**3**	**3**	**3**	**3**	**3**	**-**	**-**	**-**	**-**	**-**	**3**

Text Books:

1. Nina Godbole, Sunit Belapure, “Cyber Security”, Wiley India, New Delhi.
2. “The Indian Cyber Law” by Suresh T. Vishwanathan; Bharat Law House New Delhi.
3. “The Information technology Act, 2000”; Bare Act- Professional Book Publishers, New Delhi.

Reference Books:

1. “Cyber Law & Cyber Crimes” By Advocate Prashant Mali; Snow White Publications, Mumbai.
2. William Stallings, “Cryptography and Network Security”, Pearson Publication.

3. Kennetch J. Knapp, Cyber Security & Global Information Assurance Information Science Publishing.

Other Resources:

1. The Information Technology ACT, 2008- TIFR : https://www.tifrh.res.in
2. A Compliance Primer for IT professional: https://www.sans.org/reading-room/whitepapers/compliance/compliance-primer professionals-33538

IN-SEMESTER ASSESSMENT (50 MARKS)

1. Continuous Assessment - Theory-(20 Marks)

Suggested breakup of distribution

One MCQ test as per GATE exam pattern/ level: 05 marks.

One Class test: 05 Marks

One Open Book Test: 05 Marks

Regularity and Active participation: 05 Marks

2. Mid Semester Exam (30 Marks)

Mid semester examination will be based on 40% to 50% syllabus.

END SEMESTER EXAMINATION (50 MARKS)

End Semester Examination will be based on syllabus coverage up to the Mid Semester Examination (MSE) carrying 20%-30% weightage, and the syllabus covered from MSE to ESE carrying 70%-80% weightage.